

Lecture 4 - Density-independent stochastic growth

Announcement:

Hand back Q1

Bring laptops next class

Today's concepts:

Stochasticity (Environmental vs. Demographic) (*Won't get to considering demographic*)

Expectation vs. Variance (uncertainty)

Geometric vs. Arithmetic expectation

Review: Exponential (geometric) as first-order approximation

Show examples of population time series

Discrete vs. continuous Show videos

All previous equations have been deterministic

... if we start with same $N(0)$ and r we get exactly the same answer.

Add stochasticity: "Adding stochastic shell to deterministic core of our model"

Stochastic (from the Greek for *aim* or *guess*) = random with respect to considered variables

A stochastic process is one whose subsequent state is determined by a random element.

$$N_T = N_0 e^{(r \pm \text{noise})T} = N_0 (\lambda \pm \text{noise})^T$$

E.g., 4 years of stochastic growth

$$N_0 = 1$$

$$\lambda_1 = 2$$

$$\lambda_2 = 1$$

$$\lambda_3 = 3$$

$$\lambda_4 = 2$$

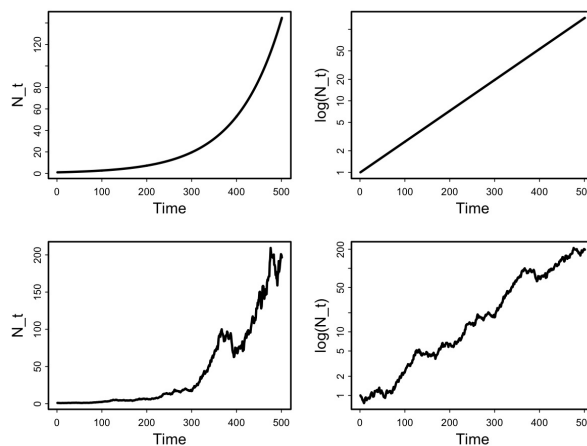
$$N_1 = N_0 \lambda_1 = 1 \cdot 2 = 2$$

$$N_2 = N_1 \lambda_2 = 2 \cdot 1 = 2$$

$$N_3 = N_2 \lambda_3 = 2 \cdot 3 = 6$$

$$N_4 = N_0 \lambda_1 \lambda_2 \lambda_3 \lambda_4 = 12$$

Show code in R...



If we wanted to determine:

Expectation (mean) of N_T ($\bar{N}_T$) and variance of N_T ($\sigma_{N_T}^2$) over all time points:

$$\bar{N}_T = \mathbb{E}_t[N_T] = \frac{1}{T} \sum_t^T N_t = \frac{1 + 2 + 2 + 6 + 12}{5} = 4.6$$

$$\begin{aligned} \sigma_{N_T}^2 &= \text{Var}[N_T] = \frac{1}{T} \sum_t^T (N_t - \bar{N}_T)^2 \\ &= \frac{(1 - 4.6)^2 + (2 - 4.6)^2 + (2 - 4.6)^2 + (6 - 4.6)^2 + (12 - 4.6)^2}{5} = 83.2/5 = 16.64 \end{aligned}$$

Note that σ = standard deviation

Environmental Stochasticity - temporal variation in population's per capita growth rate

Importance of distinguishing between geometric vs. arithmetic mean...

Example: Dynamics of two populations with same initial size:

$$N(3) = N_0 \lambda_1 \lambda_2 \lambda_3$$

$$\text{Population A: } N_A(3) = N_0 \cdot 2 \cdot 1 \cdot 3 = 6 \cdot N_0$$

$$\text{Population B: } N_B(3) = N_0 \cdot 2 \cdot 2 \cdot 2 = 8 \cdot N_0$$

$$\bar{\lambda} = \frac{1}{T} \sum_t^T \lambda_t$$

On natural (arithmetic) scale, $\bar{\lambda}_A = \bar{\lambda}_B = 2$.

So why does population B grow more?

Appropriate measure is the geometric mean (remember: popn growth is a multiplicative process):

(No standard symbol for geometric mean)

$$\text{geometric mean } \lambda = \left(\prod_t^T \lambda_t \right)^{\frac{1}{T}} = \sqrt[T]{\prod_t^T \lambda_t}$$

$$\text{Geometric mean of } \lambda_A = \sqrt[3]{\lambda_1 \cdot \lambda_2 \cdot \lambda_3} = \sqrt[3]{2 \cdot 1 \cdot 3} = 1.817...$$

$$\text{So in fact } N_A(3) = N_0 \cdot 1.817 \cdot 1.817 \cdot 1.817 = 6 \cdot N_0$$

Side note: Contrasting arithmetic vs. geometric mean

Two numbers (expressed relative to x): $\xleftarrow{a} \xrightarrow{b}$ What is x ?

$$a - x = x - b$$

$$\frac{a}{x} = \frac{x}{b}$$

$$a + b = 2x$$

$$a \cdot b = x \cdot x$$

$$\frac{a+b}{2} = x$$

$$a \cdot b = x^2$$

$$\sqrt{a \cdot b} = x$$

Will show that: *Natural log of geometric mean λ = arithmetic mean of natural log of the λ 's*

$$\text{Since... } \ln(b^a) = a \ln(b) \Rightarrow \ln \left(\left(\prod_{t=1}^T \lambda_t \right)^{1/T} \right) = \frac{1}{T} \cdot \ln(\lambda_1 \cdot \lambda_2 \dots \lambda_T)$$

$$\begin{aligned} \text{Since... } \ln(ab) = \ln(a) + \ln(b) \Rightarrow &= \frac{1}{T} (\ln(\lambda_1) + \ln(\lambda_2) + \dots + \ln(\lambda_T)) \\ &= \overline{\ln(\lambda)} \end{aligned}$$

Thus, could also calculate as:

$$\text{Geometric mean } \lambda = e^{\overline{\ln(\lambda)}}$$

An insightful approximation (provided in Case, pg. 35):

$$\text{Geometric mean } \lambda = e^{\overline{\ln(\lambda)}} \approx e^{\ln(\bar{\lambda}) - \frac{\sigma_\lambda^2}{2\bar{\lambda}^2}} = \bar{\lambda} \cdot e^{-\frac{\sigma_\lambda^2}{2\bar{\lambda}^2}}$$

$$\begin{aligned} \text{Note: } 0 < e^{-\frac{\sigma_\lambda^2}{2\bar{\lambda}^2}} < 1 \\ \text{(Note effects of } \sigma_\lambda \text{ and of } \bar{\lambda}.) \end{aligned}$$

Class R exercise - random vector draws - compare geometric and arithmetic means

- Effect of $\sigma = 0$, and increasing σ values:

Geometric mean will always be less than arithmetic mean

The more variation, the lower the geometric mean

- Effect of sample size, n :

Little effect. very low n might have more variation in depression amount

- Effect of $\bar{\lambda}$:

Raises and lowers, but note that at $\bar{\lambda} \approx 1$, geometric mean $\lambda < 1$ (declining population)!!

What does this mean for the dynamics of a given focal population?

<i>Population change</i>	Deterministic λ	Deterministic $\ln(\lambda)$	Environmental noise $\ln(\lambda \pm \text{noise})$
No change	1	0	$\ln(\bar{\lambda}) - \frac{\sigma_{\lambda}^2}{2\bar{\lambda}^2} = 0$ or NA
Growth	> 1	> 0	$\ln(\bar{\lambda}) - \frac{\sigma_{\lambda}^2}{2\bar{\lambda}^2} > 0$
Decline	< 1	< 0	$\ln(\bar{\lambda}) - \frac{\sigma_{\lambda}^2}{2\bar{\lambda}^2} < 0$

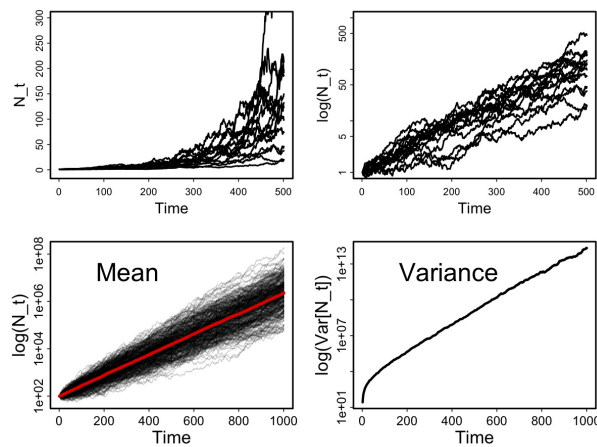
What are the expected mean and variance of N_T for an “average” (typical) population?

That is, we will now ask a different question...

⇒ What is the expected population size N_T over an ensemble of replicate populations?

(The average of n replicate populations.)

Run in R...



Will show that

$$\bar{N}_T = N_0 \bar{\lambda}^T = N_0 e^{\bar{r}T}$$

And that...

when there is *environmental variation* only:

$$\sigma_{\ln(N_T)}^2 = T \cdot \sigma_{\ln(\bar{\lambda})}^2$$

when there is only *demographic variation*:

$$\sigma_{N_T}^2 = \begin{cases} 2N_0 \bar{b}T & \text{if } \bar{b} = \bar{d} \\ \frac{\bar{b} + \bar{d}}{\bar{b} - \bar{d}} N_0 e^{\bar{r}T} (e^{\bar{r}T} - 1) & \text{if } \bar{b} \neq \bar{d} \end{cases}$$

How to get $\mathbb{E}_n[N_T]$?

Let λ be a random variable from a normal distribution.

Any given λ from this distribution is denoted by λ_t .

$$\begin{aligned}
 N_T &= N_0 \prod_{t=1}^T \lambda_t = N_0 \cdot (\lambda_1 \lambda_2 \dots \lambda_T) \\
 \mathbb{E}_n[N_T] &= N_0 \cdot \mathbb{E}_n \left[\prod_{t=1}^T \lambda_t \right] \quad (\text{since } N_0 \text{ is a constant}) \\
 &= N_0 \cdot \prod_{t=1}^T \mathbb{E}_n[\lambda] \\
 &= N_0 \cdot \mathbb{E}_n[\lambda]^T \\
 &= N_0 \cdot \bar{\lambda}^T
 \end{aligned}$$

How to get variance?

Transform by taking the log...

$$\ln(N_T) = \ln(N_0) + \ln \left(\prod_{t=1}^T \lambda_t \right) = \ln(N_0) + \boxed{\sum_{t=1}^T \ln(\lambda_t)}$$

Now working on the arithmetic scale!

Allows us to apply Central Limit Theorem (pg. 41 of Case):

The sum of independent, identically distributed random variables x_i tends to asymptote to the normal density distribution, no matter the underlying distribution of x_i

Under a sufficiently large number of independent random variable draws:

$$\begin{aligned}
 \boxed{\sum_{i=1}^n x_i} &= \mathcal{N}(n \cdot \mathbb{E}_n[x], n \cdot \text{Var}[x]) \\
 &= \mathcal{N}(n\bar{x}, n\sigma_x^2)
 \end{aligned}$$

*The mean of the sum equals the sum of all individual means
The variance of the sum equals the sum of all individual variances*

Inserting $\ln(\lambda_t)$ for x_i , we thus have

$$\begin{aligned}
 \sum_{t=1}^T \ln(\lambda_t) &= \mathcal{N}(T \cdot \mathbb{E}_n[\ln(\lambda_t)], T \cdot \text{Var}[\ln(\lambda_t)]) \\
 &= \mathcal{N}(T \cdot \overline{\ln(\lambda_t)}, T \cdot \sigma_{\ln(\bar{\lambda})}^2)
 \end{aligned}$$

Therefore, we have

$$\mathbb{E}_n[\ln(N_T)] = \ln(N_0) + T \cdot \overline{\ln(\lambda_t)}$$

and

$$\sigma_{\ln(N_T)}^2 = T \cdot \sigma_{\ln(\bar{\lambda})}^2$$

To translate expectation back to arithmetic scale:

Even though the expected value of N_T is $\mathbb{E}_n[N_T] = N_0 \cdot \bar{\lambda}^T$,

$$N_T \sim \log \mathcal{N} \text{ with median} = e^{\ln(N_0) + T \cdot \overline{\ln(\bar{\lambda})}}$$

Take-home message:

When modeling $N_{t+1} = \lambda N_t$ with environmental noise in growth rate, error must be $\log \mathcal{N}$!

In Problem Set #2 you should therefore use $N_{t+1} = N_t(\lambda \pm e^\epsilon)$, where $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$.

Summary

- The solution to $Var[N_T]$ is sensitive to assumed model-formulation!
 - Dependent on how stochasticity is assumed to affect λ .
 - If you assume $N_T = N_0(\lambda \pm \epsilon)^T$ with $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ you'll get a different prediction (and possibly negative population sizes)!
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